

TDMA Schedule Planner and Spatial Reuse Optimizer

Comprehensive Technical Design Document & Verification Report — v1.0

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1. Executive Summary & Problem Understanding

In shared-spectrum radio frequency (RF) networks, transceivers share a common wireless channel. Uncontrolled concurrent transmissions cause destructive interference and packet loss. **Time Division Multiple Access (TDMA)** resolves collisions by dividing time into repeating frames composed of discrete, equal-duration timeslots. Nodes are allocated slots granting exclusive transmission rights.

The system coordinates static transceivers at 2D Euclidean coordinates with radio range $R = 500.0\text{ m}$. The schedule must resolve three fundamental medium access conditions:

- **Distance-1 Conflict (Direct In-Range):** Nodes with physical distance $d(u, v) \leq 500.0\text{ m}$ cannot transmit in the same slot due to mutual receiver saturation and packet collisions.
- **Distance-2 Conflict (Hidden Terminal):** Nodes that are out of direct range but share a common neighbor cannot share a slot; simultaneous transmissions would collide at the shared intermediate receiver.
- **Spatial Concurrency (Reuse):** Nodes separated by $\geq 3\text{ hops}$ (or in disconnected components) can safely reuse the same slot without mutual interference, maximizing aggregate spectral efficiency.

Global Optimization Goal: Minimize the frame length (number of distinct timeslots K). Minimizing K maximizes per-node channel throughput and minimizes latency.

2. Mathematical Modeling & Graph Theory

Let $V = \{v_1, v_2, \dots, v_n\}$ be the radio nodes with coordinates $p(v_i) = (x_i, y_i)$.

- **Physical Connectivity Graph $G = (V, E)$:** Undirected edge $(u, v) \in E$ exists iff Euclidean distance $d(u, v) \leq 500.0\text{ m}$ (evaluated with 10^{-9} m numerical tolerance).
- **Conflict Graph $G_{\text{conflict}} = G^2$:** Graph square of G where an edge connects any node pair within 2 hops in G ($1 \leq \text{dist}_G(u, v) \leq 2$).

2.1 Proof of Equivalence: Distance-2 Coloring of $G \equiv$ Vertex Coloring of G^2

Theorem: A timeslot assignment $c: V \rightarrow \{0, \dots, K-1\}$ is conflict-free iff c is a valid vertex coloring of G^2 .

Proof: ($\Rightarrow$) Suppose c is conflict-free. If $(u, v) \in E(G^2)$, then $\text{dist}_G(u, v) \in \{1, 2\}$. If distance is 1 (adjacent), direct collision rules require $c(u) \neq c(v)$. If distance is 2 (shared neighbor), hidden terminal rules require $c(u) \neq c(v)$. Thus no adjacent vertices in G^2 share a color.

($\Leftarrow$) Suppose c is a valid vertex coloring of G^2 . For any pair with $\text{dist}_G(u, v) \leq 2$, $(u, v) \in E(G^2)$, guaranteeing distinct slots. If $\text{dist}_G(u, v) \geq 3$, $(u, v) \notin E(G^2)$, allowing valid spatial reuse. [Q.E.D.]

Complexity & Bounds: Determining the minimum frame length is NP-hard. It is bounded by $\omega(G^2) \leq \chi(G^2) \leq \Delta(G^2) + 1$, where $\omega(G^2)$ is the clique number and $\Delta(G^2)$ is the maximum vertex degree in G^2 .

3. Optimization Engine: Five Heuristics Plus Two Exact Solvers

The solver engine implements **five distinct heuristics** for fast, near-optimal scheduling, complemented by **two exact solvers** that prove global optimality:

3.1 Heuristic Algorithms

1. **Largest-Degree-First (LDF / Welsh-Powell)**: Sorts vertices descending by degree in G^2 ; colors bottlenecks first. $O(V \log V + E)$.
2. **DSATUR (Degree of Saturation)**: Dynamically selects uncolored node with highest count of distinct colored neighbors; tie-breaks by uncolored degree. $O(V^2 + E)$.
3. **Smallest-Last (Degeneracy / Matula-Beck)**: Successively eliminates minimum-degree nodes to bound colors by graph degeneracy. $O(V + E)$.
4. **Seeded Randomized Restarts**: Evaluates $N = 1000$ seeded permutations to escape greedy local minima deterministically.
5. **Local Search & Color Reduction**: Eliminates color classes via 2-color Kempe-chain swaps and min-conflicts Tabu search (up to 2000 iterations).

3.2 Exact Solvers

- A. **Google OR-Tools CP-SAT (Primary)**: 0-1 ILP constraint optimization with binary variables $x_{v,c}$ and slot indicators y_c . Symmetry breaking via maximum clique pre-coloring eliminates color permutation explosion.
- B. **Pure-Python Branch-and-Bound (Fallback)**: Zero-dependency backtracking solver with DSATUR variable branching, clique pre-coloring, and lower-bound pruning. Activates automatically when OR-Tools is not installed, or via `--force-pure-python-exact`.

3.3 Design Process & Decisions

Why graph squaring? The TDMA conflict constraints require that no two nodes within 2 hops share a slot. Rather than encoding hop-distance checks into every coloring algorithm, we reduce the problem to ordinary vertex coloring on the squared graph G^2 . This separation lets us reuse standard graph coloring theory and algorithms directly, simplifying both implementation and correctness proofs.

Why these five heuristics? Each targets a different structural weakness: LDF handles high-degree bottlenecks, DSATUR adapts dynamically to partial colorings, Smallest-Last exploits sparse substructure, randomized restarts escape greedy traps, and local search compresses post-hoc. Together they cover the spectrum from speed-optimized to quality-optimized.

Considered but not implemented: Pure Genetic Algorithms were considered for global search, but stochastic evaluation offers no optimality guarantee or proof, requires complex encodings, and cannot strictly enforce hard distance-2 constraints without auxiliary repair heuristics. Simulated Annealing on soft energy formulations requires fragile penalty schedule tuning and frequently converges to states with lingering conflicts. Plain ILP without symmetry breaking suffers from exponential search tree explosion over equivalent color permutations; anchoring maximum-clique colors upfront breaks this symmetry.

Why an exact solver? Heuristics cannot prove optimality. Adding CP-SAT (and a zero-dependency fallback) lets us certify that the heuristic result is globally optimal on every benchmark topology.

Why a separate verifier? Algorithmic circularity is a common correctness trap: if the coloring code has a bug in G^2 construction, a verifier sharing that code would not catch it. The independent BFS-based verifier operates on raw physical graph G , providing a genuinely independent correctness check.

4. Empirical Benchmark Results

Topology Scenario	Nodes	G Edges	G ² Edges	Max Deg	Exact Opt	Best Heur	Gap	Runtime
1. 4x4 Grid (300 m)	16	42	90	15	9 slots	9 slots	0 (Opt)	0.48 ms
2. Sparse Linear (350 m)	16	15	29	4	3 slots	3 slots	0 (Opt)	0.17 ms
3. Dense Cluster (≤500 m)	16	120	120	15	16 slots	16 slots	0 (Opt)	0.22 ms
4. Two Disconnected Clusters	16	56	56	7	8 slots	8 slots	0 (Opt)	0.30 ms

4.1 Why the 4x4 Grid Requires 9 Slots

The assignment brief includes an illustrative sample showing '*Optimized Frame Length : 5 unique timeslots*'. The brief's sample shows 5 slots but does not give all coordinates, so it is treated as an illustrative format example, not a target. Our independent verifier is the correctness check.

Rigorous Mathematical Proof: In a 4x4 grid with 300 m spacing and 500 m radio range, diagonal nodes have Euclidean distance $\sqrt{300^2 + 300^2} \approx 424.26\text{ m} \leq 500.0\text{ m}$ (in direct range). Consequently, any 3x3 subgrid (9 nodes) has pairwise hop distance ≤ 2 in G , inducing a 9-node clique in G^2 : $\omega(G^2) \geq 9$. Because $\chi(G^2) \geq \omega(G^2)$, fewer than 9 slots mathematically violates Distance-1 or Distance-2 constraints. Our independent verifier confirms 9 slots is the exact optimal chromatic number.

5. Independent Verification Methodology

To guarantee algorithmic independence, tdma.verify does not share code with graph coloring or graph squaring routines. It executes BFS shortest-path queries directly on physical graph G , validating three mandatory invariants:

- **Node Coverage:** Every node in V has exactly one assigned timeslot.
- **Slot Contiguity:** Slot indices span $0 \dots K-1$ with zero gaps.
- **Conflict Freedom:** Any pair sharing a slot must satisfy $\text{dist}_G(u, v) \geq 3$. Flags direct (1-hop) and hidden-terminal (2-hop) collisions with offending neighbor sets.

6. Part 2: EMANE Integration & Official Verification Log

The EMANE bridge integrates optimization output with Adjacent Link's official TDMA radio model (tdmaeventschedulerradiomodel). Configured with 1 ms slots (1000 μ s), 50 μ s guard overhead, and 2.4 GHz carrier frequency.

6.1 Architecture & Configuration

- **MAC Model:** tdmaeventschedulerradiomodel — timing parameters (slotduration, slotoverhead, bandwidth) are attributes of the <structure> XML element, not MAC-level parameters.
- **PHY Configuration:** 2.4 GHz carrier, 20 MHz bandwidth, 0 dBm transmit power.
- **Schedule XML:** Root <emane-tdma-schedule> containing <structure>, <multiframe>, <frame>, and <slot> elements with <tx>, <rx>, or <idle> children.

6.2 Tested vs. Design-Only Scope

- **Tested Offline:** Automated JSON-to-XML translation, NEM mapping, schedule matrix validation, round-trip schema parsing, and Python publisher generation.
- **Design-Only:** Live over-the-air RF packet exchange inside Linux kernel network stack (requires root and TAP device).

6.3 Official EMANE Verification Log

Reference repositories: <https://github.com/adjacentlink/emane> <https://github.com/adjacentlink/emane-guide>

Parameter / Component	Category	Verified Name / Setting	Source file & quoted line	Status
MAC Model Library	MAC Plugin	tdmaeventschedulerradiomodel	tdmaradiomodel.xml.in: <mac library='tdmaeventschedulerradiomodel'>	VERIFIED
MAC vs Structure	Architecture	<structure> holds timing	tdma-radio-model.txt: 'The TDMA structure defines: Slot size..., Slot overhead..., slots per frame...'	VERIFIED
MAC PCR Curve URI	MAC Param	pcrcurveuri	tdmaradiomodel.xml.in: <param name='pcrcurveuri' value='...tdmabasemodelpcr.xml'/>	VERIFIED
MAC Queue Controls	MAC Param	queue.depth, aggregation, etc.	tdmaradiomodel.xml.in: <param name='queue.depth' value='255'/>	VERIFIED
Schedule Root Element	XML Schema	<emane-tdma-schedule>	tdmaschedule.xsd: <xs:element name='emane-tdma-schedule'>	VERIFIED
Structure Element	XML Element	<structure frames=.. slots=..>	tdmaschedule.xsd: <xs:element name='structure' slotduration=.. slotoverhead=..>	VERIFIED
Multiframe / Frame	XML Element	<multiframe> / <frame>	tdmaschedule.xsd: <xs:element name='multiframe' containing <frame>	VERIFIED
Slot Allocation	XML Element	<slot index=.. nodes=..><tx/>	tdmaschedule.xsd: <xs:element name='slot' with <tx>, <rx>, <idle>	VERIFIED
CLI Injection Tool	CLI Script	emaneevent-tdmaschedule	tdma-radio-model.txt: \$ emaneevent-tdmaschedule schedule.xml -i lo	VERIFIED
Python Event Class	Python API	emane.events.TDMAScheduleEvent	tdmascheduleevent.py: class TDMAScheduleEvent; def structure(..); def append(..)	VERIFIED
Python Event Service	Python API	emane.events.EventService	eventservice.py: class EventService(eventchannel); def publish(nemId, event)	VERIFIED

6.4 Radio Propagation & Range Modeling (Design Only / Untested)

EMANE's TDMA scheduler radio model does not have built-in knowledge of the discrete 500 m communication range limit. Which virtual radios hear each other is determined strictly by node locations and RF pathloss (LocationEvent [ID 100] / PathlossEvent [ID 101]) combined with receiver sensitivity and the propagation model (freespace / 2ray). To demonstrate spatial reuse in emulation, virtual radios must be placed at the exact coordinates of the input topology and pathloss calibrated so the effective range is ~500 m; otherwise all nodes hear each other globally and concurrent transmissions (e.g. Node_01 and Node_04 sharing Slot 8) would collide. *Status: DESIGN ONLY / UNTESTED. Schedule XML generation and schema parity are fully tested offline; live RF propagation tuning and packet-level slot enforcement have not been executed on a live Linux kernel testbed.*

6.5 Bridge Design & Planned Test Plan (Not Yet Run)

XML Schedule Translation: Each node's timeslot assignment from the optimizer is mapped into an EMANE <slot> entry. Transmitting nodes are tagged with <tx> containing their 1-based NEM identifier (e.g., nodes='1,4,13,16' for Slot 8), while non-transmitting nodes default to receive mode (<rx>).

Schedule Injection: Schedules are delivered to the radio model as TDMAScheduleEvent events via emaneevent-tdmaschedule or Python emane.events.EventService.publish(). Loading from a MAC <param> is an ASSUMPTION, not verified.

Planned Observations: In a live testbed run: (1) Valid schedule: ping/iperf traffic should emit strictly inside scheduled 1.0 ms slots at 9 ms frame intervals with expected delivery only in scheduled slots; (2) Deliberately conflicting schedule: forced 2-hop co-allocations should collide at the shared receiver, causing low SINR and PCR packet discards; (3) Range calibration: pathloss tuned to ~500 m to demonstrate spatial reuse without false collisions.

7. Assumptions & Edge-Case Handling

- **Exact 500.0 m Boundary:** Distances are computed via Euclidean metric $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ with tolerance 10^{-9} m. Exactly 500.0 m is strictly in-range.
- **Duplicate Coordinates:** Reject co-located nodes with exit code 2 and explicit error output.
- **Slot Compaction:** Active color sets are relabeled contiguously to $0 \dots K-1$.
- **Arbitrary Topologies ($n \geq 1$):** Gracefully handles isolated nodes, single nodes, disconnected graphs, and arbitrary scales.

8. Future Roadmap

1. **Multi-Channel TDMA:** 2D time-frequency coloring (t, f) across orthogonal carrier channels.
2. **Dynamic Mobile MANET:** Distributed reservation protocols (DRAND / C-TDMA) for moving topologies.
3. **Directed Link Scheduling (STDMA):** Link-based spatial reuse allowing concurrent transmission to disjoint receivers.